ALGEBRA 2 · UNIT 10: TRIGONOMETRIC FUNCTIONS

Ferris Wheel & Tides

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students model the height of a Ferris wheel car and the depth of a harbor with sinusoidal functions. They use radians and the unit circle to place cars, find amplitude, midline, period, and phase shift, solve trig equations to find when a car is high or a boat can dock, and design their own ride.

Periodic motion gives meaning to every parameter of a sinusoid. Starting at the bottom makes the reflected cosine natural, and tide schedules turn trig equations into docking windows.

Standards Addressed

- F-TF.1–2 Radians and the unit circle
- F-TF.5 Model periodic phenomena with trig functions
- F-IF.7e Graph trig functions showing period, midline, amplitude
- F-TF.7 Solve trig equations in context (Honors)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a Ferris wheel video. Part 1 unit circle and parameters.
Day 2	Part 2 equations and timing.
Day 3	Part 3 design.

Materials

- Student handout
- Graphing calculators (radian mode) or Desmos
- Paper plates for a unit-circle model (optional)

How the Three Versions Differ

Level	What changes
CP2	Radians and car positions, a table and graph, reading amplitude, midline, and period.
CP1	Write equations, solve for times above a height, and model tides with docking windows.
Honors	Phase shifts and equivalent forms, inverse trig solutions, and a design meeting timing constraints.

Watch For (Common Misconceptions)

- Calculator in degree mode.
- Using the period as B.
- Using the diameter as the amplitude.

Extension Ideas

- Use real tide tables from NOAA.
- Model daylight hours over a year.
- Film a bike wheel and model a point on the tire.

Answer Key – CP2

Part 1: The Wheel

#	Task	Answer
1	How many degrees apart are the cars? How many radians?	360 ÷ 16 = 22.5°; ⅛π radians
2	Convert 135° and 270° to radians.	¾π; 1½π (3π ÷ 2)
3	Find the amplitude (radius), midline (center height), and period of the ride.	Amplitude 60 ft, midline 70 ft, period 4 min
4	The model is h(t) = 70 − 60 cos(½πt). Complete the table.	10, 70, 130, 70, 10

Part 2: Tides

#	Task	Answer
5	Find the maximum depth, minimum depth, and period.	Max 14 ft, min 6 ft, period 12 hours
6	At what times is the tide highest and lowest during the first 12 hours?	Highest at 3 a.m., lowest at 9 a.m.
7	Sketch one cycle of the tide.	Starts at 10 ft, up to 14 at t = 3, back to 10 at t = 6, down to 6 at t = 9, back to 10 at t = 12.

Part 3: Design a Ride

#	Task	Answer
8	Choose a diameter, center height, and time for one turn.	Answers vary.
9	Write h(t) = (center) − (radius) cos(2πt ÷ period).	Answers vary.
10	Make a table for one turn and sketch the graph.	Answers vary.

Answer Key – CP1

Part 1: The Wheel

#	Task	Answer
1	Write h(t) and explain each number. Why is B = ½π?	h(t) = 70 − 60 cos(½πt): midline 70, amplitude 60, minus sign because the ride starts at the bottom, B = 2π ÷ 4 = ½π.
2	Find h(0.5) and h(1.5) exactly.	h(0.5) = 70 − 60 cos(¼π) = 70 − 30√2 ≈ 27.6 ft; h(1.5) = 70 − 60 cos(¾π) = 70 + 30√2 ≈ 112.4 ft
3	For how long during each turn is a rider above 100 ft? Solve 70 − 60 cos(½πt) = 100.	cos(½πt) = −½ → ½πt = ⅔π or 1⅓π → t = 1⅓ or 2⅔ min. Above 100 ft for 1⅓ min (80 s).

Part 2: Tides

#	Task	Answer
4	A boat needs at least 8 ft of water. Solve d(t) = 8 for 0 ≤ t ≤ 12.	sin(⅙πt) = −½ → ⅙πt = 7π ÷ 6 or 11π ÷ 6 → t = 7 or t = 11
5	When can the boat use the harbor during the first 12 hours?	From midnight to 7 a.m. and from 11 a.m. to noon (d ≥ 8)
6	Rewrite the model if high tide were at 5 a.m. instead (same amplitude, midline, and period).	d(t) = 4 sin(⅙π(t − 2)) + 10 (or 4 cos(⅙π(t − 5)) + 10)

Part 3: Design a Ride

#	Task	Answer
7	Write your model.	Answers vary.
8	Solve for the time above 80% of the maximum height.	Answers vary.
9	Present the design.	Answers vary.

Answer Key – Honors

Part 1: Equivalent Models

#	Task	Answer
1	Show that $70 - 60\cos(\frac{1}{2}\pi t) = 70 + 60\sin(\frac{1}{2}\pi(t - 1))$.	$\sin(\theta - \frac{1}{2}\pi) = -\cos\theta$, so $60\sin(\frac{1}{2}\pi(t - 1)) = -60\cos(\frac{1}{2}\pi t)$.
2	A rider boards from a platform at the 3 o'clock position (70 ft) moving up. Write the model.	$h(t) = 70 + 60\sin(\frac{1}{2}\pi t)$
3	Find all times in the first 8 minutes when the bottom-boarding rider is at 40 ft.	$\cos(\frac{1}{2}\pi t) = 0.5 \rightarrow \frac{1}{2}\pi t = \frac{1}{3}\pi + 2\pi k$ or $1\frac{2}{3}\pi + 2\pi k \rightarrow t = \frac{2}{3}, 3\frac{1}{3}, 4\frac{2}{3}, 7\frac{1}{3}$ min

Part 2: Harbor Planning

#	Task	Answer
4	Write a model for the depth.	Midline 8, amplitude 5, $B = 2\pi \div 12.4$: $d(t) = 8 - 5\cos(2\pi(t - 2) \div 12.4)$, t hours after midnight
5	A ship needs 10 ft. Find the first window after 2 a.m. when it can enter (to the nearest minute).	$\cos(2\pi(t - 2) \div 12.4) = -0.4 \rightarrow 2\pi(t - 2) \div 12.4 \approx 1.982 \rightarrow t - 2 \approx 3.91$ h. From about 5:55 a.m. until about $2 + 12.4 - 3.91 \approx$ 10:29 a.m.
6	Explain why the docking windows shift later each day.	Each cycle is 12.4 hours, so two cycles take 24.8 hours, about 48 minutes more than a day.

Part 3: Design Within Constraints

#	Task	Answer
7	Choose parameters and write the model.	Answers vary.
8	Prove that your design meets every constraint.	Answers vary.
9	Present it.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Designed ride's function matches its description, and timing answers are found by solving trig equations. Goes beyond the minimum with a creative, realistic choice.	Designed ride's function matches its description, and timing answers are found by solving trig equations.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes both models with amplitude, midline, and period explained, a graph, and solved timing questions. Polished, organized, and visually engaging.	Report includes both models with amplitude, midline, and period explained, a graph, and solved timing questions. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.